

Advanced Lesson 12 - Exercises: Pure Functions

Mikey Ferguson

MATH 7013 - Advanced Mathematical Modeling

1: Applying pure functions to values

```
Clear[x, f, data, newData];
```

```
(#12 &)[3]
```

```
(Sin[#1] &)[ $\frac{\pi}{2}$ ]
```

```
({#1, Sin[#1]} &)[ $\frac{\pi}{2}$ ]
```

```
({#1, Sin[#2]} &)[1,  $\frac{\pi}{2}$ ]
```

```
(#1 ≥ 10 &)[12]
```

```
(#1 ≥ 10 &)[8]
```

```
(#1 ≥ #2 &)[12, 8]
```

```
(#1 ≥ #2 &)[8, 12]
```

2: Selecting the points at most 20

```
data = {4, 65, 12, 4, 8, 21, 4};
```

```
Select[data, #1 ≤ 20 &]
```

3: Selecting the points between 5 and 20 inclusive

```
Select[data, 5 ≤ #1 ≤ 20 &]
```

4: Mapping to ordered pairs

```
Clear[f];  
({#1, f[#1]} &) /@ data
```

5: Selecting points whose first coordinate squared is at most the second

```
newData = {{1, 4}, {3, 2}, {2, 2}, {5, 1}, {2, 5}};  
Select[newData, #1[[1]]2 ≤ #1[[2]] &]
```

6: Derivatives as pure functions

```
Clear[f, x];  
f'[x]  
f'  
Sin'[x]  
Sin'
```